

Glossary

Advanced Data Analytics



Terms and definitions from all courses

A

A/B testing: A way to compare two versions of something to find out which version performs better

Absolute values: (Refer to **observed values**)

Accuracy: Refers to the proportion of data points that were correctly categorized

Action: A Tableau tool to help an audience interact with a visualization or dashboard by allowing control of selection

Active listening: Refers to allowing team members, bosses, and other collaborative stakeholders to share their own points of view before offering responses

AdaBoost: (Refer to **adaptive boosting**)

Adaptive boosting: A boosting methodology where each consecutive base learner assigns greater weight to the observations incorrectly predicted by the preceding learner

Addition rule (for mutually exclusive events): The concept that if the events A and B are mutually exclusive, then the probability of A or B happening is the sum of the probabilities of A and B

Adjusted R^2 : A variation of R^2 that accounts for having multiple independent variables present in a linear regression model

Affinity: The metric used to calculate the distance between points/clusters

agg(): A pandas groupby method that allows the user to apply multiple calculations to groups of data

Agglomerative clustering: A clustering methodology that works by first assigning every point to its own cluster, then progressively combining clusters based on intercluster distance

Aggregate information: Data from a significant number of users that has eliminated personal information

Algorithm: A set of instructions for solving a problem or accomplishing a task

Aliasing: A process that allows the user to assign an alternate name—or alias—to something

Analysis of Variance (ANOVA): A group of statistical techniques that test the difference of means between three or more groups

Analytics Team Manager: A data professional who supervises analytical strategy for an organization, often managing multiple groups

Analyze stage: Stage of the PACE workflow where the necessary data is acquired from primary and secondary sources and then cleaned, reorganized, and analyzed

ANCOVA (Analysis of Covariance): A statistical technique that tests the difference of means between three or more groups while controlling for the effects of covariates, or variable(s) irrelevant to the test

append(): A method that adds an element to the end of a list

Array: An ordered collection of items of a single data type

Array(): A function for converting input to an array

Argument: Information given to a function in its parentheses

Artificial intelligence (AI): Refers to computer systems able to perform tasks that normally require human intelligence

AI tool: AI-powered software that can automate or assist users with a variety of tasks

AI prompt library: A personalizable Google Sheet that contains all of the prompts included in this course to support you during your job search

AI user: Someone who leverages AI to complete a personal or professional task

Assignment: The process of storing a value in a variable

Attribute: A value associated with an object or class which is referenced by name using dot notation

Average: The distance between each cluster's centroid and other clusters' centroids

B

Backward elimination: A stepwise variable selection process that begins with the full model, with all possible independent variables, and removes the independent variable that adds the least explanatory power to the model

Bagging: A technique used by certain kinds of models that use ensembles of base learners to make predictions; refers to the combination of bootstrapping and aggregating

Base learner: Each individual model that comprises an ensemble

Bayes' rule: (Refer to **Bayes' theorem**)

Bayes' theorem: An equation that can be used to calculate the probability of an outcome or class, given the values of predictor variables

Bayesian inference: (Refer to **Bayesian statistics**)

Bayesian statistics: A powerful method for analyzing and interpreting data in modern data analytics; also referred to as Bayesian inference

Behavioral interview: A type of interview designed to understand how you responded in past situations to predict your future performance

Best fit line: The line that fits the data best by minimizing some loss function or error

Bias: In data structuring, refers to organizing data results in groupings, categories, or variables that are misrepresentative of the whole dataset

Bias-variance trade-off: Balance between two model qualities, bias and variance, to minimize overall error for unobserved data

Bin: A segment of data that groups values into categories

Binning: Grouping continuous values into a smaller number of categories, or intervals

Binomial distribution: A discrete distribution that models the probability of events with only two possible outcomes: success or failure

Binomial logistic regression: A technique that models the probability of an observation falling into one of two categories, based on one or more independent variables

Binomial logistic regression linearity assumption: An assumption stating that there should be a linear relationship between each X variable and the logit of the probability that Y equals one

Black-box model: Any model whose predictions cannot be precisely explained

Boolean: A data type that has only two possible values, usually true or false

Boolean data: A data type that has only two possible values, usually true or false

Boolean masking: A filtering technique that overlays a Boolean grid onto a dataframe in order to select only the values in the dataframe that align with the True values of the grid

Boosting: A technique that builds an ensemble of weak learners sequentially, with each consecutive learner trying to correct the errors of the one that preceded it

Bootstrapping: Refers to sampling with replacement

Box plot: A data visualization that depicts the locality, spread, and skew of groups of values within quartiles

Branching: The ability of a program to alter its execution sequence

break: A keyword that lets a user escape a loop without triggering any ELSE statement that follows it in the loop

Business Intelligence Analyst: (Refer to **Business Intelligence Engineer**)

Business Intelligence Engineer: A data professional who uses their knowledge of business trends and databases to organize information and make it accessible; also referred to as a Business Intelligence Analyst

C

Career Dreamer: An experimental tool that uses AI to make career exploration easier and more personalized

Career identity statement: A few sentences about yourself that provide a quick but insightful overview of your key skills, interests, career goals, and what you're passionate about

Case interview: An interview type designed to evaluate your problem-solving, analytical, and communication skills by presenting you with real-world business challenges to analyze and solve

Categorical data: Data that is divided into a limited number of qualitative groups

Categorical variables: Variables that contain a finite number of groups or categories

Causation: Describes a cause-and-effect relationship where one variable directly causes the other to change in a particular way

Cells: The modular code input and output fields into which Jupyter Notebooks are partitioned

Central Limit Theorem: The idea that the sampling distribution of the mean approaches a normal distribution as the sample size increases

Centroid: The center of a cluster determined by the mathematical mean of all the points in that cluster

Chat: An input box in an AI tool where you can add prompts

Chi-squared (χ^2) Goodness of Fit Test: A hypothesis test that determines whether an observed categorical variable follows an expected distribution

Chi-squared (χ^2) Test for Independence: A hypothesis test that determines whether or not two categorical variables are associated with each other

Chief Data Officer: An executive-level data professional who is responsible for the consistency, accuracy, relevancy, interpretability, and reliability of the data a team provides

Child node: A node that is pointed to from another node

Class imbalance: When a dataset has a predictor variable that contains more instances of one outcome than another

Class: An object's data type that bundles data and functionality together

Classical probability: A type of probability based on formal reasoning about events with equally likely outcomes

Cleaning: The process of removing errors that might distort your data or make it less useful; one of the six practices of EDA

Cluster random sample: A probability sampling method that divides a population into clusters, randomly selects certain clusters, and includes all members from the chosen clusters in the sample

Collaborative filtering: A technique used by recommendation systems to make comparisons based on who else liked the content

Collective outliers: A group of abnormal points, following similar patterns and isolated from the rest of the population

Comparator: An operator that compares two values and produces Boolean values (True/False)

Complement of an event: In statistics, refers to an event not occurring

Complement rule: A concept stating that the probability that event A does not occur is one minus the probability of A

Complete: The maximum pairwise distance between clusters

Composition: Refers to defining attributes and methods at the instance level to have a more differentiated relationship between objects in the same class

Computer programming: The process of giving instructions to a computer to perform an action or set of actions

concat(): A pandas function that combines data either by adding it horizontally as new columns for existing rows or vertically as new rows for existing columns

Concatenate: To link or join together

Concatenation: Refers to building longer strings out of smaller strings

Conditional probability: Refers to the probability of an event occurring given that another event has already occurred

Conditional statement: A section of code that directs the execution of programs

Confidence band: The area surrounding a line that describes the uncertainty around the predicted outcome at every value of X

Confidence interval: A range of values that describes the uncertainty surrounding an estimate

Confidence level: A measure that expresses the uncertainty of the estimation process

Confusion matrix: A graphical representation of how accurate a classifier is at predicting the labels for a categorical variable

Construct stage: Stage of the PACE workflow where data models and machine learning algorithms are built, interpreted, and revised to uncover relationships within the data and help unlock insights from those relationships

Constructor: A special method to add values to an instance in object creation

Content-based filtering: A technique used by recommendation systems to make comparisons based on attributes of content

Contextual outliers: Normal data points under certain conditions but become anomalies under most other conditions

Continuous random variable: A variable that takes all the possible values in some range of numbers

Continuous: A mathematical concept indicating that a measure or dimension has an infinite and uncountable number of outcomes

Continuous variables: Variables that can take on an infinite and uncountable set of values

Convenience sample: A non-probability sampling method that involves choosing members of a population that are easy to contact or reach

Correlation: Measures the way two variables tend to change together

Cover letter: A personalized document that showcases your communication skills and personality while demonstrating how your skills and experience align with the employer's needs

Cross-validation: A process that uses different portions of the data to test and train a model on different iterations

CSV file: A plaintext file that uses commas to separate distinct values from one another; Stands for "comma-separated values"

Customer churn: The business term that describes how many and at what rate customers stop using a product or service, or stop doing business with a company

D

Data anonymization: The process of protecting people's private or sensitive data by eliminating PII

Data cleaning: The process of formatting data and removing unwanted material

Data engineer: A data professional who makes data accessible, ensures data ecosystems offer reliable results, and manages infrastructure for data across enterprises

Data ethics: Well-founded standards of right and wrong that dictate how data is collected, shared, and used

Data governance: A process for ensuring the formal management of a company's data assets

Data professional: Any individual who works with data and/or has data skills

Data science: The discipline of making data useful

Data scientist: A data professional who works closely with analytics to provide meaningful insights that help improve current business operations

Data source: The location where data originates

Data stewardship: The practices of an organization that ensures that data is accessible, usable, and safe

Data structure: A collection of data values or objects that contain different data types

Data type: An attribute that describes a piece of data based on its values, its programming language, or the operations it can perform

DataFrame: A two-dimensional, labeled data structure with rows and columns

Data visualization: A graph, chart, diagram, or dashboard that is created as a representation of information

Database (DB) file: A file type used to store data, often in tables, indexes, or fields

Dataframe: A two-dimensional data-structure organized into rows and columns

DBSCAN: A clustering methodology that searches data space for continuous regions of high density; stands for “density-based spatial clustering of applications with noise”

Debugging: Troubleshooting, or searching for errors in a script or program

Decision node: A node of the tree where decisions are made

Decision tree: A flowchart-like structure that uses branching paths to predict the outcomes of events, or the probability of certain outcomes

Deduplication: The elimination or removal of matching data values in a dataset

def: A keyword that defines a function at the start of the function block

Dependent events: The concept that two events are dependent if one event changes the probability of the other event

Dependent variable (Y): The variable a given model estimates

Describe(): A function that returns the statistical summary of a dataframe or series, including mean, standard deviation, and minimum and maximum column values.

Descriptive statistics: A type of statistics that summarizes the main features of a dataset

dict(): A function used to create a dictionary

Dictionary: A data structure that consists of a collection of key-value pairs

difference(): A function that finds the elements present in one set but not the other

Dimensions: Qualitative data values used to categorize and group data to reveal details about it

Discovering: The process data professionals use to familiarize themselves with the data so they can start conceptualizing how to use it; one of the six practices of EDA

Discrete features: Features with a countable number of values between any two values

Discrete random variable: A variable that has a countable number of possible values

Discrete: A mathematical concept indicating that a measure or dimension has a finite and countable number of outcomes

distance_threshold: A hyperparameter in agglomerative clustering models that determines the distance above which clusters will not be merged

Documentation: An in-depth guide that is written by the developers who created a package that features very specific information on various functions and features

Documentation string: A group of text that explains what a method or function does; also referred to as a “docstring”

Dot notation: How to access the methods and attributes that belong to an instance of a class

Downsampling: The process of removing some observations from the majority class, making it so they make up a smaller percentage of the dataset than before

Dummy variables: Variables with values of 0 or 1 that indicate the presence or absence of something

dtype: A NumPy attribute used to check the data type of the contents of an array

Dynamic typing: Variables that can point to objects of any data type

Dynamic value: A value the user inputs or the output of a program, an operation, or a function

E

Econometrics: A branch of economics that uses statistics to analyze economic problems

Edge computing: A way of distributing computational tasks over a bunch of nearby processors (i.e., computers) that is good for speed and resiliency and does not depend on a single source of computational power

elif: A reserved keyword that executes subsequent conditions when the previous conditions are not true

else: A reserved keyword that executes when preceding conditions evaluate as False

Empirical probability: A type of probability based on experimental or historical data

Empirical rule: A concept stating that the values on a normal curve are distributed in a regular pattern, based on their distance from the mean

Ensemble learning: Refers to building multiple models and aggregating their predictions

Ensembling: (Refer to **ensemble learning**)

Enumerate(): A built-in function that iterates through a sequence and tracks each element and its place in the index

eps (Epsilon): In DBSCAN clustering models, a hyperparameter that determines the radius of a search area from any given point

Errors: In a regression model, the natural noise assumed to be in a model

Escape character: A character that changes the typical behavior of the characters that follow it

Execute stage: Stage of the PACE workflow where a data professional will present findings with internal and external stakeholders, answer questions, consider different viewpoints, and make recommendations

Explanatory variable: (Refer to **independent variable**)

Explicit conversion: The process of converting a data type of an object to a required data type

Exploratory data analysis (EDA): The process of investigating, organizing, and analyzing datasets and summarizing their main characteristics, often by employing data wrangling and visualization methods; the six main practices of EDA are: discovering, structuring, cleaning, joining, validating, and presenting

Expression: A combination of numbers, symbols, or other variables that produce a result when evaluated

Extra Sum of Squares F-test: Quantifies the difference between the amount of variance that is left unexplained by a reduced model that is explained by the full model

Extracting: The process of retrieving data out of data sources for further data processing

Extrapolation: A model's ability to predict new values that fall outside of the range of values in the training data

F

F1-Score: The harmonic mean of precision and recall

False positive: A test result that indicates something is present when it really is not

Feature engineering: The process of using practical, statistical, and data science knowledge to select, transform, or extract characteristics, properties, and attributes from raw data

Feature extraction: A type of feature engineering that involves taking multiple features to create a new one that would improve the accuracy of the algorithm

Feature selection: A type of feature engineering that involves selecting the features in the data that contribute the most to predicting the response variable

Feature transformation: A type of feature engineering that involves modifying existing features in a way that improves accuracy when training the model

Filtering: The process of selecting a smaller part of a dataset based on specified values and using it for viewing or analysis

First-party data: Data that was gathered from inside your own organization

Float: A data type that represents numbers that contain decimals

For loop: A piece of code that iterates over a sequence of values

format(): A string method that formats and inserts specific substrings into designated places within a larger string

Forward selection: A stepwise variable selection process that begins with the null model—with zero independent variables—and considers all possible variables to add; incorporates the independent variable that contributes the most explanatory power to the model

Function: A body of reusable code for performing specific processes or tasks

G

Gemini: A generative AI model that can understand and combine various types of information, including text, code, audio, images, and video

Gemini Deep Research: A feature within the Gemini apps that allows users to conduct in-depth research by browsing numerous websites, analyzing the information, and generating multi-page reports

Gemini Live: A feature in the Gemini app that allows for natural, real-time voice conversations

Generative AI: AI that can generate new content, like text, images, or other media

Generator(): A function that returns an object (iterator) which can be iterated over (one value at a time)

Global outliers: Values that are completely different from the overall data group and have no association with any other outliers

Global variable: A variable that can be accessed from anywhere in a program or script

Gradient boosting machines (GBMs): Model ensembles that use gradient boosting

Gradient boosting: A boosting methodology where each base learner in the sequence is built to predict the residual errors of the model that preceded it

GridSearch: A tool to confirm that a model achieves its intended purpose by systematically checking every combination of hyperparameters to identify which set produces the best results, based on the selected metric

groupby(): A pandas DataFrame method that groups rows of the dataframe together based on their values at one or more columns, which allows further analysis of the groups

Grouping: The process of aggregating individual observations of a variable into groups

Growth mindset: A commitment to ongoing learning and development

H

Hackathon: An event where programmers and data professionals come together and work on a project

Head(): A function that returns a preview of the column names and the first few rows of a dataset

Heatmap: A type of data visualization that depicts the magnitude of an instance or set of values based on two colors

Help(): A Python help function used to display the documentation of modules, functions, classes, keywords, and more

Histogram: A data visualization that depicts an approximate representation of the distribution of values in a dataset

Hold-out sample: A random sample of observed data that is not used to fit the model

Homoscedasticity assumption: An assumption of simple linear regression stating that the variation of the residuals (errors) is constant or similar across the model

Human touch: The act of personalizing or refining AI output to ensure the personality, tone, voice, language use, and accuracy are as authentic as possible

Hyperparameters: Parameters that can be set by the modeler before the model is trained

Hyperparameter tuning: Refers to changing parameters that directly affect how the model trains, before the learning process begins

Hypothesis: A theory or an explanation, based on evidence, that has not yet been refuted

Hypothesis testing: A statistical procedure that uses sample data to evaluate an assumption about a population parameter

I

if: A reserved keyword that sets up a condition in Python

iloc[]: A type of notation in pandas that indicates when the user wants to select by integer-location-based position

Immutability: The concept that a data structure or element's values can never be altered or updated

Immutable data type: A data type in which the values can never be altered or updated

Implicit conversion: The process Python uses to automatically convert one data type to another without user involvement

Import statement: A statement that uses the import keyword to load an external library, package, module, or function into the computing environment

Independent events: The concept that two events are independent if the occurrence of one event does not change the probability of the other event

Independent observation assumption: An assumption of simple linear regression stating that each observation in the dataset is independent

Independent variable (X): The variable whose trends are associated with the dependent variable

index(): A string method that outputs the index number of a character in a string

Indexing: A way to refer to the individual items within an iterable by their relative position

Inertia: The sum of the squared distances between each observation and its nearest centroid

Inferential statistics: A type of statistics that uses sample data to draw conclusions about a larger population

Info(): Gives the total number of entries, along with the data types—called Dtypes in pandas—of the individual entries

Inheritance: Refers to letting a programmer build relationships between concepts and group them together to reduce code duplication

Inner join: A way of combining data such that only the keys that are in both dataframes get included in the merge

Input validation: The practice of thoroughly analyzing and double-checking to make sure data is complete, error-free, and high-quality

Input: Information entered into a program

input(): A Python function that can be used to ask a question in a message and store the answer in a variable

insert(): A function that takes an index as the first parameter and an element as the second parameter, then inserts the element into a list at the given index

Instance variable: A variable that is declared in a class outside of other methods or blocks

Instantiation: Refers to creating a copy of the class that inherits all class variables and methods

Int64: A standard integer data type, representing numbers somewhere between negative nine quintillion and positive nine quintillion

Integer: A data type used to represent whole numbers without fractions

Integrated Development Environment (IDE): A piece of software that has an interface to write, run, and test a piece of code

Interaction term: Represents how the relationship between two independent variables is associated with changes in the mean of the dependent variable

Intercept (constant B_0): The y value of the point on the regression line where it intersects with the y-axis

Interpersonal skills: Traits that focus on communicating and building relationships

Interquartile range: The distance between the first quartile (Q1) and the third quartile (Q3)

intersection(): A function that finds the elements that two sets have in common

Interval: A sample statistic plus or minus the margin of error

Interval estimate: A calculation that uses a range of values to estimate a population parameter

Interview: A meeting designed to assess a candidate's skills, experience, and fit for a role through a structured conversation involving questions and answers

is: A rule that checks objects and classes for ancestry

items(): A dictionary method to retrieve both the dictionary's keys and values

Iterable: An object that's looped, or iterated, over

Iterate: To repeat a process or set of instructions, often with the aim of improvement or reaching a desired outcome

Iteration: The repeated execution of a set of statements, where one iteration is the single execution of a block of code

J

Job search plan: A document or checklist that outlines the steps you need to take to effectively manage your job search

Job search platforms: Websites used to search for and apply for jobs

Job search portfolio: A folder you create in your Google Drive for all of the materials you create in this course to keep you organized and prepared throughout the job search process

Job search tracker: A system to organize application information and materials

Joining: The process of augmenting data by adding values from other datasets; one of the six practices of EDA

JSON file: A data storage file that is saved in a JavaScript format

Jupyter Notebook: An open-source web application for creating and sharing documents containing live code, mathematical formulas, visualizations, and text

K

K-means: An unsupervised partitioning algorithm used to organize unlabeled data into groups, or clusters

Kernel: An underlying core program, like Python

Keys: The shared points of reference between different dataframes

keys(): A dictionary method to retrieve only the dictionary's keys

Keyword: A special word in a programming language that is reserved for a specific purpose and that can only be used for that purpose

L

Label encoding: Data transformation technique where each category is assigned a unique number instead of a qualitative value

Large Language Model (LLM): A type of AI algorithm that uses deep learning techniques to identify patterns in text and map how different words and phrases relate to each other

Leaf node: The nodes where a final prediction is made

learning_rate: In XGBoost, a hyperparameter that specifies how much weight is given to each consecutive tree's prediction in the final ensemble

Left join: A way of combining data such that all of the keys in the left dataframe are included, even if they aren't in the right dataframe

Len(): A function used to measure the length of strings

Library: A reusable collection of code; also referred to as a "package"

Likelihood: The probability of observing the actual data, given some set of beta parameters

Line: A collection of an infinite number of points extending in two opposite directions

Linear regression: A technique that estimates the linear relationship between a continuous dependent variable and one or more independent variables

Linearity assumption: An assumption of simple linear regression stating that each predictor variable (X_i) is linearly related to the outcome variable (Y)

Link function: A nonlinear function that connects or links the dependent variable to the independent variables mathematically

Linkage: The method used to determine which points/clusters to merge

List: A data structure that helps store and manipulate an ordered collection of items

List comprehension: Formulaic creation of a new list based on the values in an existing list

Literacy rate: The percentage of the population in a given age group that can read and write

loc[]: Notation that is used to select pandas rows and columns by name

Log-Odds function: (Refer to **logit**)

Logical operator: An operator that connects multiple statements together and performs complex comparisons

Logistic regression: A technique that models a categorical dependent variable (Y) based on one or more independent variables (X)

Logit: The logarithm of the odds of a given probability

Loop: A block of code used to carry out iterations

Loss function: A function that measures the distance between the observed values and the model's estimated values

Lower limit: When constructing an interval, the calculation of the sample means minus the margin of error

M

Machine learning: The use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data

MAE (Mean Absolute Error): The average of the absolute difference between the predicted and actual values

Magic commands: Commands that are built into IPython to simplify common tasks

Magics: (Refer to **magic commands**)

MANCOVA (Multivariate Analysis of Covariance): An extension of ANCOVA and MANOVA that compares how two or more continuous outcome variables vary according to categorical independent variables, while controlling for covariates

MANOVA (Multivariate Analysis of Variance): An extension of ANOVA that compares how two or more continuous outcome variables vary according to categorical independent variables

Margin of error: The maximum expected difference between a population parameter and a sample estimate

Markdown: A markup language that lets the user write formatted text in a coding environment or plain-text editor

matplotlib: A library for creating static, animated, and interactive visualizations in Python

max_depth: In tree-based models, a hyperparameter that controls how deep each base learner tree will grow

max_features: In decision tree and random forest models, a hyperparameter that specifies the number of features that each tree randomly selects during training called "colsample_bytree" in XGBoost

Maximum Likelihood

Estimation (MLE): A technique for estimating the beta parameters that maximizes the likelihood of the model producing the observed data

Mean: The

average value in a dataset

Measure of central tendency: A value that represents the center of a dataset

Measure of dispersion: A value that represents the spread of a dataset, or the amount of variation in data points

Measure of position: A method by which the position of a value in relation to other values in a dataset is determined

Measures: Numeric values that can be aggregated or placed in calculations

Median: The middle value in a dataset

Mentor: Someone who shares knowledge, skills, and experience to help another grow both professionally and personally

merge(): A pandas function that joins two dataframes together; it only combines data by extending along axis one horizontally

Merging: A method to combine two (or more) different dataframes along a specified starting column(s)

Method: A function that belongs to a class and typically performs an action or operation

Metrics: Methods and criteria used to evaluate data

min_child_weight: In XGBoost models, a hyperparameter indicating that a tree will not split a node if it results in any child node with less weight than this value called “min_samples_leaf” in decision tree and random forest models

min_samples: In DBSCAN clustering models, a hyperparameter that specifies the number of samples in an ϵ -neighborhood for a point to be considered a core point (including itself)

min_samples_leaf: In decision tree and random forest models, a hyperparameter that defines the minimum number of samples for a leaf node called “min_child_weight” in XGBoost

min_samples_split: In decision tree and random forest models, a hyperparameter that defines the minimum number of samples that a node must have to split into more nodes

Missing data: A data value that is not stored for a variable in the observation of interest

Mode: The most frequently occurring value in a dataset

Model assumptions: Statements about the data that must be true in order to justify the use of a particular modeling technique

Model selection: The process of determining which model should be the final product and put into production

Model validation: The set of processes and activities intended to verify that models are performing as expected

Modularity: The ability to write code in separate components that work together and that can be reused for other programs

Module: A simple Python file containing a collection of functions and global variables

Modulo: An operator that returns the remainder when one number is divided by another

MSE (Mean Squared Error): The average of the squared difference between the predicted and actual values

Multiple linear regression: A technique that estimates the relationship between one continuous dependent variable and two or more independent variables

Multiple regression: (Refer to multiple linear regression)

Multiplication rule (for independent events): The concept that if the events A and B are independent, then the probability of both A and B happening is the probability of A multiplied by the probability of B

Mutability: The ability to change the internal state of a data structure

Mutually exclusive: The concept that two events are mutually exclusive if they cannot occur at the same time

N

n_clusters: In K-means and agglomerative clustering models, a hyperparameter that specifies the number of clusters in the final model

N-dimensional array: The core data object of NumPy; also referred to as “ndarray”

n_estimators: In random forest and XGBoost models, a hyperparameter that specifies the number of trees your model will build in its ensemble

Naive Bayes: A supervised classification technique that is based on Bayes’s Theorem with an assumption of independence among predictors

Naming conventions: Consistent guidelines that describe the content, creation date, and version of a file in its name

Naming restrictions: Rules built into the syntax of a programming language

NaN: How null values are represented in pandas; stands for “not a number”

ndim: A NumPy attribute used to check the number of dimensions of an array

Negative correlation: An inverse relationship between two variables, where when one variable increases, the other variable tends to decrease, and vice versa

Nested loop: A loop inside of another loop

Networking: The process of building authentic professional relationships, making true connections that last and are mutually rewarding

No multicollinearity assumption: An assumption of simple linear regression stating that no two independent variables (X_i and X_j) can be highly correlated with each other

Non-null count: The total number of data entries for a data column that are not blank

Non-probability sampling: A sampling method that is based on convenience or the personal preferences of the researcher, rather than random selection

None: A special data type in Python used to indicate that things are empty or that they return nothing

Nonprofit: A group organized for purposes other than generating profit; often aims to further a social cause or provide a benefit to the public

Nonresponse bias: Refers to when certain groups of people are less likely to provide responses

Normal distribution: A continuous probability distribution that is symmetrical on both sides of the mean and bell-shaped

Normality assumption: An assumption of simple linear regression stating that the residuals are normally distributed

NotebookLM: A personalized AI research assistant that answers questions and generates insights based on the sources you upload

NumPy: An essential library that contains multidimensional array and matrix data structures and functions to manipulate them



Object: A collection of data that consists of variables and methods or functions

Object type: A component category, usually associated with its respective class

Object-oriented programming: A programming system that is based around objects which can contain both data and code that manipulates that data

Objective probability: A type of probability based on statistics, experiments, and mathematical measurements

Observed values: The existing sample of data, where each data point in the sample is represented by an observed value of the dependent variable and an observed value of the independent variable

One hot encoding: A data transformation technique that turns one categorical variable into several binary variables

One-Way ANOVA: A type of statistical testing that compares the means of one continuous dependent variable based on three or more groups of one categorical variable

Online presence: Your online presence is your personal brand across all of your online activities

Open data: Data that is available to the public and free to use, with guidance on how to navigate the datasets and acknowledge the source

Ordinary least squares estimation (OLS): A common way to calculate linear regression coefficients

Outcome variable (Y): (Refer to dependent variable)

Outer join: A way of combining data such that all of the keys from both dataframes get included in the merge

Outliers: Observations that are an abnormal distance from other values or an overall pattern in a data population

Output: (1) A message stating what to do next; (2) The result produced by prompting an AI tool to generate text or an image or complete some other task

Overfitting: When a model fits the observed or training data too specifically and is unable to generate suitable estimates for the general population

P

P-value: The probability of observing results as extreme as those observed when the null hypothesis is true

PACE: A workflow data professionals can use to remain focused on the end goal of any given dataset; stands for plan, analyze, construct, and execute

Package: A fundamental unit of shareable code that others have developed for a specific purpose

pandas: A powerful library built on top of NumPy that's used to manipulate and analyze tabular data

Panel interview: An interview type where the candidate is asked questions by a group of people at the same time, often used to draw upon multiple perspectives and increase the likelihood of hiring the best candidate

Parameter: A characteristic of a population

Percentile: The value below which a percentage of data falls

Personally identifiable information (PII): Information that permits the identity of an individual to be inferred by either direct or indirect means

Plan stage: Stage of the PACE workflow where the scope of a project is defined and the informational needs of the organization are identified

Point estimate: A calculation that uses a single value to estimate a population parameter

Poisson distribution: A probability distribution that models the probability that a certain number of events will occur during a specific time period

pop(): A method that extracts an element from a list by removing it at a given index

Popularity bias: The phenomenon of more popular items being recommended too frequently

Population: Every possible element that a data professional is interested in measuring

Population proportion: The percentage of individuals or elements in a population that share a certain characteristic

Positive correlation: A relationship between two variables that tend to increase or decrease together.

Post hoc test: An ANOVA test that performs a pairwise comparison between all available groups while controlling for the error rate

Posterior probability: The probability of an event occurring after taking into consideration new information

Precision: The proportion of positive predictions that were correct to all positive predictions

Predicted values: The estimated Y values for each X calculated by a model

Predictor variable: (Refer to independent variable)

Presenting: The process of making a cleaned dataset available to others for analysis or further modeling; one of the six practices of EDA

Prior probability: Refers to the probability of an event before new data is collected

Probability: The branch of mathematics that deals with measuring and quantifying uncertainty

Probability distribution: A function that describes the likelihood of the possible outcomes of a random event

Probability sampling: A sampling method that uses random selection to generate a sample

Program: A series of instructions written so that a computer can perform a certain task, independent of any other application

Programming languages: The words and symbols used to write instructions for computers to follow

Professional brand: The professional image and reputation you cultivate through your online presence, networking interactions, communication style, and personal presentation

Professional pitch: A concise summary of your background, skills, and career aspirations useful for introductions at networking events and informational interviews

Prompt: Text input that provides instructions to the AI model on how to generate output

Purposive sample: A method of non-probability sampling that involves researchers selecting participants based on the purpose of their study

Python: A general-purpose programming language

Q

Quartile: A value that divides a dataset into four equal parts

R

R² (The Coefficient of Determination): Measures the proportion of variation in the dependent variable, Y, explained by the independent variable(s), X

RACI chart: A visual that helps to define roles and responsibilities for individuals or teams to ensure work gets done efficiently; lists who is responsible, accountable, consulted, and informed for project tasks

Random experiment: A process whose outcome cannot be predicted with certainty

Random forest: An ensemble of decision trees trained on bootstrapped data with randomly selected features

Random seed: A starting point for generating random numbers

Random variable: A variable that represents the values for the possible outcomes of a random event

Range: The difference between the largest and smallest value in a dataset

range(): A Python function that returns a sequence of numbers starting from zero, increments

by 1 by default, and stops before the given number

Recall: The proportion of actual positives that were identified correctly to all actual positives

Recommendation systems: Unsupervised learning techniques that use unlabeled data to offer relevant suggestions to users

Refactoring: The process of restructuring code while maintaining its original functionality

Regression analysis: A group of statistical techniques that use existing data to estimate the relationships between a single dependent variable and one or more independent variables

Regression coefficient: The estimated betas in a regression model

Regression models: (Refer to **regression analysis**)

Regularization: A set of regression techniques that shrinks regression coefficient estimates towards zero, adding in bias, to reduce variance

remove(): A method that removes an element from a list

Representative sample: A sample that accurately reflects the characteristics of a population

reshape(): A NumPy method used to change the shape of an array

Residual: The difference between observed or actual values and the predicted values of the regression line

Resilience: The ability to maintain motivation and emotional well-being, persevere through setbacks, bounce back from rejections, leverage past experiences, and maintain a positive outlook throughout the job search process

Response variable: (Refer to **dependent variable**)

Responsible AI: The principle of developing and using AI ethically, with the intent of benefitting people and society while avoiding harm

Resume: A comprehensive document used to communicate your relevant skills and experience to potential employers

return: A reserved keyword in Python that makes a function produce new results which are saved for later use

Reusability: The capability to define code once and using it many times without having to rewrite it

Right join: A way of combining data such that all the keys in the right dataframe are included—even if they aren't in the left dataframe

Root node: The first node of the tree, where the first decision is made

S

Sample: A segment of a population that is representative of the entire population

Sample size: The number of individuals or items chosen for a study or experiment

Sample space: The set of all possible values for a random variable

Sampling: The process of selecting a subset of data from a population

Sampling bias: Refers to when a sample is not representative of the population as a whole

Sampling distribution: A probability distribution of a sample statistic

Sampling frame: A list of all the items in a target population

Sampling variability: Refers to how much an estimate varies between samples

Sampling with replacement: Refers to when a population element can be selected more than one time

Sampling without replacement: Refers to when a population element can be selected only one time

Scatterplot matrix: A series of scatterplots that show the relationships between pairs of variables

Script: A collection of commands in a file designed to be executed like a program

Seaborn: A visualization library based on matplotlib that provides a simpler interface for working with common plots and graphs

Second-party data: Data that was gathered outside your organization directly from the original source

Self: A parameter passed to a method or attributes used to instantiate an object

Self-documenting code: Code written in a way that is readable and makes its purpose clear

Semantics: Refers to the variables and objects that give meaning to Python code

Sequence: A positionally-ordered collection of items

Series: A one-dimensional labeled array capable of holding any data type

Set: A data structure in Python that contains only unordered, non-interchangeable elements; a Tableau term for a custom field of data created from a larger dataset based on custom conditions

Set(): A function that takes an iterable as an argument and returns a new set object

shape: A NumPy attribute used to check the shape of an array

Shrinkage: (Refer to **learning_rate**)

Simple random sample: A probability sampling method in which every member of a population is selected randomly and has an equal chance of being chosen

Silhouette analysis: The comparison of different models' silhouette scores

Silhouette score: The mean of the silhouette coefficients of all the observations in a model

Simple linear regression: A technique that estimates the linear relationship between one independent variable, X, and one continuous dependent variable, Y

Simple random sample: A probability sampling method in which every member of a population is selected randomly and has an equal chance of being chosen

Single: The minimum pairwise distance between clusters

Slicing: A method for breaking information down into smaller parts to facilitate efficient examination and analysis from different viewpoints

Slope: The amount that y increases or decreases per one-unit increase of x

Snowball sample: A method of non-probability sampling that involves researchers recruiting initial participants to be in a study and then asking them to recruit other people to participate in the study

Sorting: The process of arranging data into a meaningful order for analysis

Standard deviation: A statistic that calculates the typical distance of a data point from the mean of a dataset

Standard error: The standard deviation of a sample statistic

Standard error of the mean: The sample standard deviation divided by the square root of the sample size

Standard error of the proportion: The square root of the sample proportion times one minus the sample proportion divided by the sample size

Standardization: The process of putting different variables on the same scale

STAR method: A storytelling framework that helps you communicate your experiences and skills in a clear, concise, and compelling way to help an interviewer understand exactly how you successfully managed certain situations; stands for Situation, Task, Action, and Result

Statistic: A characteristic of a sample

Statistical significance: The claim that the results of a test or experiment are not explainable by chance alone

Statistics: The study of the collection, analysis, and interpretation of data

Story: A Tableau term for a group of dashboards or worksheets assembled into a presentation

Stratified random sample: A probability sampling method that divides a population into groups and randomly selects some members from each group to be in the sample

String: A sequence of characters and punctuation that contains textual information

String literal: A programming string used in code in which characters exist as the value themselves, rather than as variables

String slice: The portion of a string that can contain more than one character; also referred to as a substring

Structuring: The process of taking raw data and organizing or transforming it to be more easily visualized, explained, or modeled; one of the six practices of EDA

Subjective probability: A type of probability based on personal feelings, experience, or judgment

Sum of squared residuals (SSR): The sum of the squared difference between each observed value and its associated predicted value

Summary statistics: A method that summarizes data using a single number

Supervised machine learning: A category of machine learning that uses labeled datasets to train algorithms to classify or predict outcomes

Supervised model: A machine learning model that is used to make predictions about unseen events

symmetric_difference(): A function that finds elements from both sets that are mutually not present in the other

Syntax: The structure of code words, symbols, placement, and punctuation

Systematic random sample: A probability sampling method that puts every member of a population into an ordered sequence, chooses a random starting point in the sequence, and selects members for the sample at regular intervals

T

Tableau: A business intelligence and analytics platform that helps people visualize, understand, and make decisions with data

Tabular data: Data that is in the form of a table, with rows and columns

Target population: The complete set of elements that someone is interested in knowing more about

Technical interview: An interview type designed to assess your technical competency; common in fields like computer engineering, computer science, and IT

Third-party data: Data gathered outside your organization and aggregated

tolist(): A NumPy method to convert arrays into lists

Transferable skills: Skills you've learned in one situation that can be applied to a completely different situation, like a new job

Tree-based learning: A type of supervised machine learning that performs classification and regression tasks

Tuple: An immutable sequence that can contain elements of any data type

tuple(): A function that transforms input into tuples

Two-Way ANOVA: A type of statistical testing that compares the means of one continuous dependent variable based on three or more groups of two categorical variables

type(): A function used to identify the type of data in a list

U

Undercoverage bias: Refers to when some members of a population are inadequately represented in a sample

union(): A function that finds all the elements from both sets

Unsupervised model: A machine learning model that is used to discover the natural structure of the data, finding relationships within unlabeled data

Upper limit: When constructing an interval, the calculation of the sample means plus the margin of error

Upsampling: The process of taking observations from the minority class and either adding copies of those observations to the dataset or generating new observations to add to the dataset

V

Validating: The process of verifying that the data is consistent and high quality; one of the six practices of EDA

values(): A dictionary method to retrieve only the dictionary's values

Variable: A named container which stores values in a reserved location in the computer's memory

Variable selection: The process of determining which variables or features to include in a given model

Variance inflation factors (VIF): Quantifies how correlated each independent variable is with all of the other independent variables

Variance: Refers to model flexibility and complexity, so the model learns from existing data; the average of the squared difference of each data point from the mean

Vectorization: A process that enables operations to be performed on multiple components of a data object at the same time

Voluntary response sample: A method of non-probability sampling that consists of members of a population who volunteer to participate in a study

W

Ward: Merges two clusters whose merging will result in the lowest inertia

Weak

learner: A model that performs slightly better than randomly guessing

While

loop: A loop that instructs the computer to continuously execute the code based on the value of a condition

X

XGBoost (extreme gradient boosting): An optimized GBM package

X-Y-Z approach: A formula or technique that helps you structure your accomplishments to clearly showcase your impact, especially in an interview or job application materials; X= accomplishment or action; Y = quantifiable result; Z = method or skills used

Z

Zero Frequency problem: Occurs when the dataset has no occurrences of a class label and some value of a predictor variable together

Z-score: A measure of how many standard deviations below or above the population mean a data point is